1 (a) Simplify $x^8 \times x^3$

x^{11}

(1)

(b) Simplify $(5y)^3$

$5y \times 5y \times 5y$

$125y^3$

(1)

(c) Simplify $\frac{w^7}{w^4}$

w^3

(1)

(Total for question 1 is 3 marks)

---

2 (a) Simplify $a^9 \times a^4$

a^{13}

(1)

(b) Simplify $(4b^2c)^3$

$4b^2c \times 4b^2c \times 4b^2c$

$64b^6c^3$

(2)

(c) Simplify $d^9 \div d^4$

d^5

(1)

(Total for question 2 is 4 marks)

---

3 (a) Simplify $2m^2 \times 5n^6$

$10m^2n^6$

(1)

(b) Simplify $15p^3 \div 3p^4$

$5p^{-1}$

(2)

(Total for question 3 is 3 marks)

4 (a) Simplify $(t^3)^4$

t^{12}

(1)

(b) Simplify $12m^2n^6 \div 3mn^4$

$4mn^2$

(2)

(Total for question 4 is 3 marks)

---

5 Simplify $5m^2n^3 \times 3mn^4$

$15m^3n^7$

(Total for question 5 is 2 marks)

---

6 (a) Write down the value of 5^{-3}

$\frac{1}{125}$

(1)

(b) Write down the value of 5^0

1

(1)

(Total for question 6 is 2 marks)

---

7 Work out the value of $5^2 \times 2^3$

25×8

200

(Total for question 7 is 1 mark)

8 Write down the value of 2^{-3}

$\frac{1}{8}$

(Total for question 8 is 1 mark)

---

9 $y^2 \times y^a = y^7$

(a) Find the value of a.

5

(1)

$(y^4)^b = y^{12}$

(b) Find the value of b.

3

(1)

(Total for question 9 is 2 marks)

---

10 (a) Given $\frac{x^6}{x^a} = x^8$

Find the value of a.

$a =$ -2

(1)

(b) Simplify $(2m^2)^4$

$16m^8$

(2)

(Total for question 10 is 3 marks)

11 (a) Write $\frac{3^4 \times 3^5}{3^2}$ as a power of 3

$$\frac{3^9}{3^2} = 3^7$$

3^7

(2)

(b) Write down the value of 3^{-3}

$\frac{1}{27}$

(1)

(c) Write down the value of 3^0

1

(1)

(Total for question 11 is 4 marks)

---

12 Work out the value of $\frac{2^9 \times 2^{-2}}{2^4}$

$$\frac{2^7}{2^4} = 2^3 = 8$$

$\cancel{2^3}$ 8

(Total for question 12 is 2 marks)

---

13 Work out the value of $(2^2)^3$

$$4^3 = 64$$

64

(Total for question 13 is 1 mark)

14 (a) Simplify $p^3 \times p^5$

p^8

(1)

(b) Simplify $(4ab^2)^3$

$64a^3b^6$

(2)

(c) Simplify $\dfrac{16m^7n^3}{4m^3n}$

$4m^4n^2$

(2)

(Total for question 14 is 5 marks)

---

15 $1000^4 = 10^x$

Find the value of x.

$(10^3)^4 = 10^x$

$10^{12} = 10^x$

12

(Total for question 15 is 1 mark)

---

16 Work out the value of $\dfrac{2^3 \times 2}{2^5}$

$\dfrac{2^4}{2^5} = 2^{-1} = \dfrac{1}{2}$

$\dfrac{1}{2}$

(Total for question 16 is 2 marks)

---

17 Write down the reciprocal of 8

$\dfrac{1}{8}$

(Total for question 17 is 1 mark)

18 (a) Simplify $9p^3 \times 2p^{-2}$

$18p$

(1)

(b) Simplify $(5x^3y^2)^3$

$125x^9y^6$

(2)

(c) $p^3 \times p^5 = p^{12} \times p^y$

Find the value of y

$p^8 = p^{12} \times p^y$

-4

(2)

(Total for question 18 is 5 marks)

---

19 $10^x = 1$

Write down the value of x.

0

(Total for question 19 is 1 mark)

---

20 Write $5^4 \times 5$ as a power of 5

5^5

(Total for question 20 is 1 mark)

---

21 Write down the reciprocal of 2

$\frac{1}{2}$

(Total for question 21 is 1 mark)

22 (a) Simplify $5c^2d^3 \times 2d$

$10c^2d^4$

(1)

(b) Write 64×4^5 as a power of 4

$4^3 \times 4^5 = 4^8$

4^8

(2)

(c) Simplify $p^3 \times (p^5)^2$

$p^3 \times p^{10}$

p^{13}

(2)

(Total for question 22 is 5 marks)

---

23 $p^9 \times p^5 = p^x$

Write down the value of x

14

(Total for question 23 is 1 mark)

---

24 Write down the reciprocal of $\frac{1}{3}$

3

(Total for question 24 is 1 mark)

---

25 Simplify $\frac{10p^3q^5r}{4p^3q^6}$

$\frac{\cancel{10}^{5}\cancel{p^3}\cancel{q^5}r}{\cancel{4}_{2}\cancel{p^3}q^{\cancel{6}}}$ $\frac{5r}{2q}$

$\frac{5r}{2q}$

(Total for question 25 is 2 marks)

OR $2.5rq^{-1}$